

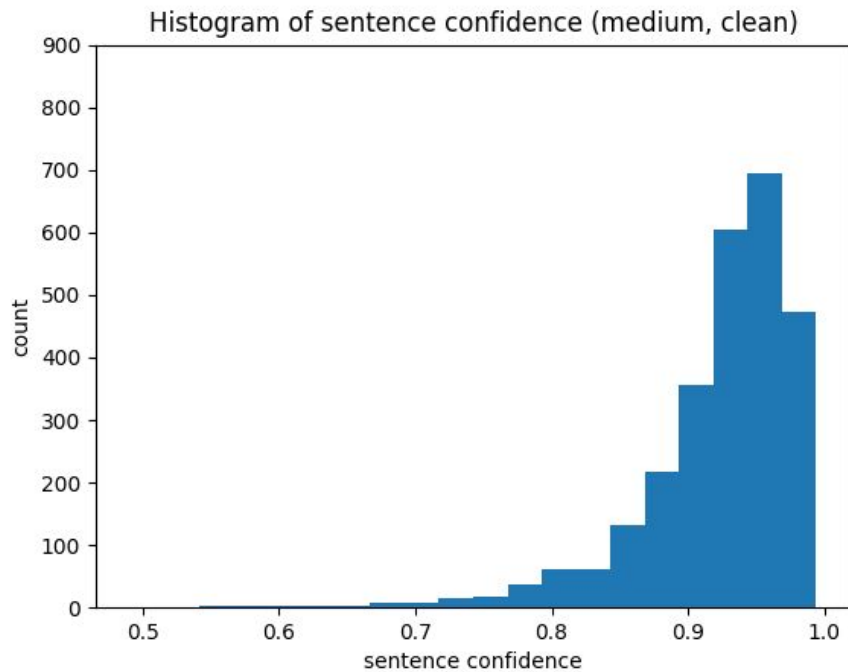
Experiments with Medium whisper model (clean)

Maryam Naderi

Sentence confidence

First we start with sentence confidence level in medium whisper model and also clean Librispeech.

Histogram



Let's look at the results for GPT-3.5-turbo first:

We have two models here: gpt-3.5-turbo-1106 and gpt-3.5-turbo-0125

gpt-3.5-turbo-0125

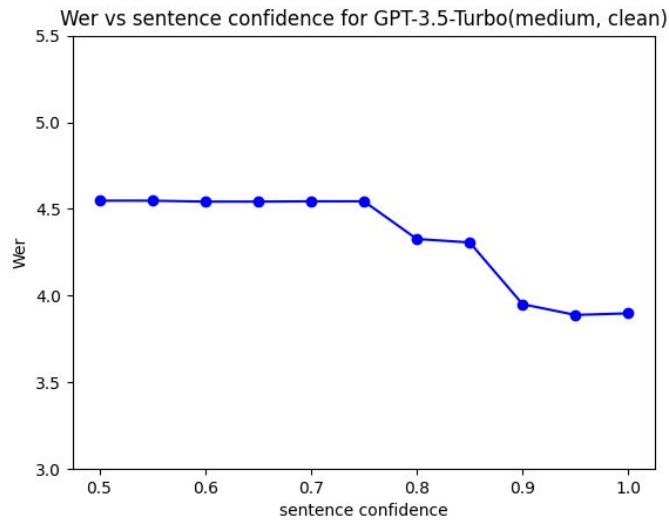
```
Number of evaluated audio files: 2702
WER_original: 4.5477%
WER_corrected_chatgpt: 3.8982%
CER_original: 1.9548%
CER_corrected_chatgpt: 1.5181%
SER_original: 39.0081%
SER_corrected_chatgpt: 38.1569%
---
```

gpt-3.5-turbo-1106

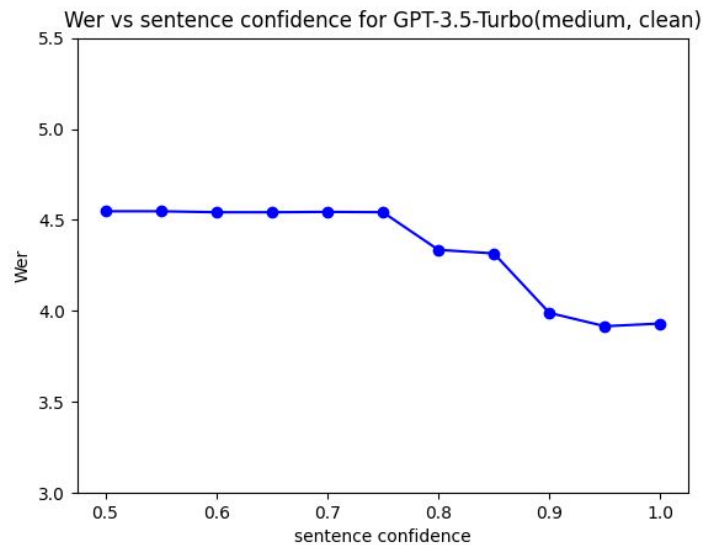
```
Number of evaluated audio files: 2702
WER_original: 4.5477%
WER_corrected_chatgpt: 3.9310%
CER_original: 1.9548%
CER_corrected_chatgpt: 1.5561%
SER_original: 39.0081%
SER_corrected_chatgpt: 38.3050%
---
```

WER vs sentence confidence

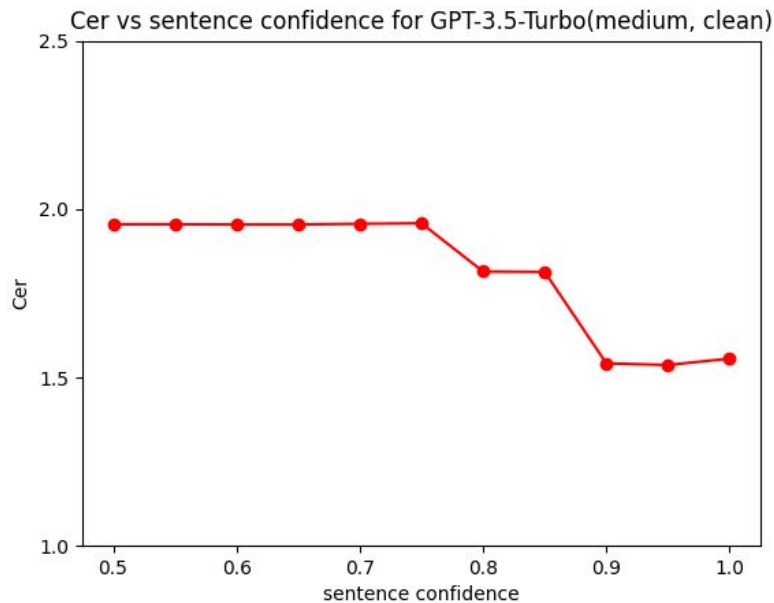
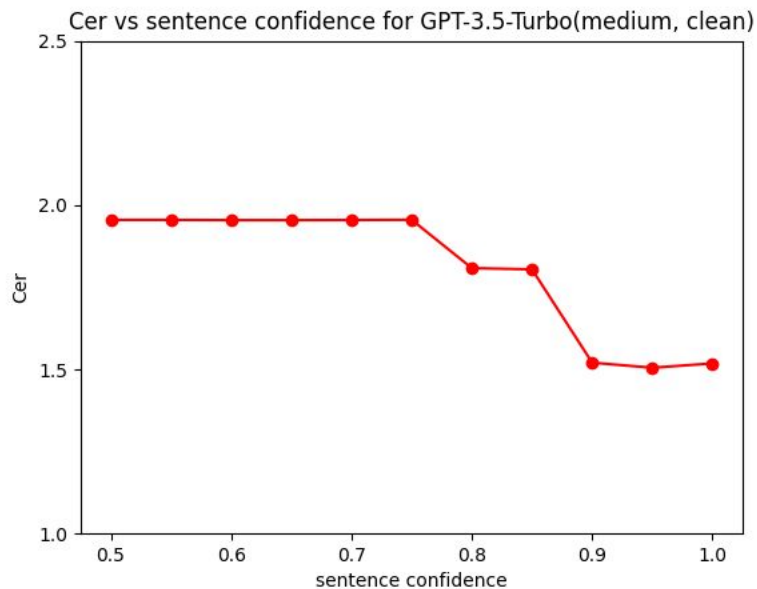
gpt-3.5-turbo-0125



gpt-3.5-turbo-1106



CER vs sentence confidence



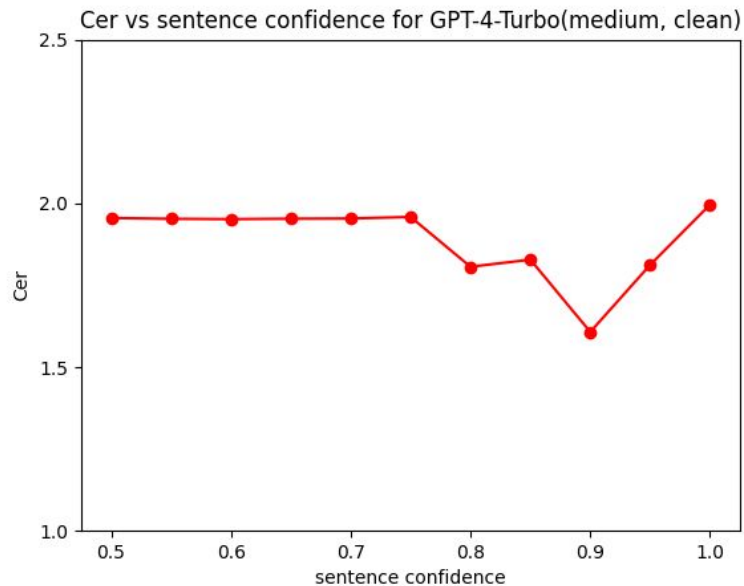
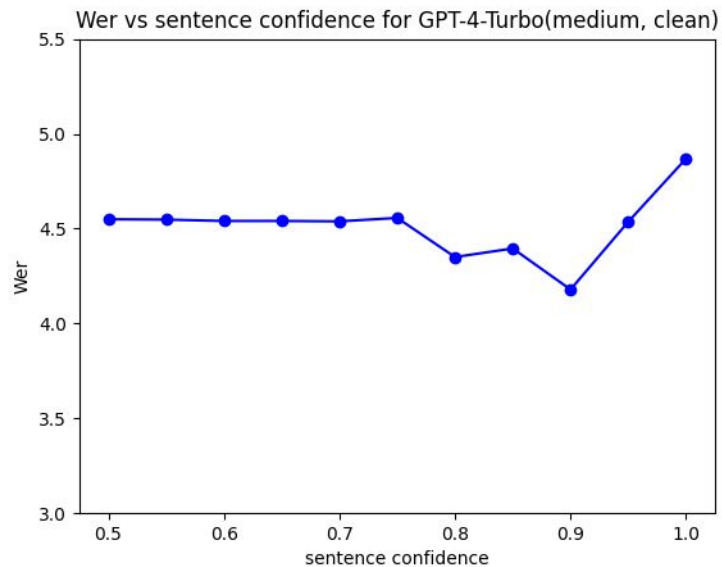
Now let's look at the results for GPT-4-turbo:

We have one models here: gpt-4-1106-preview

Results for gpt-4-turbo

```
Number of evaluated audio files: 2702
WER_original: 4.5477%
WER_corrected_chatgpt: 4.8694%
CER_original: 1.9548%
CER_corrected_chatgpt: 1.9960%
SER_original: 39.0081%
SER_corrected_chatgpt: 47.4463%
---
```

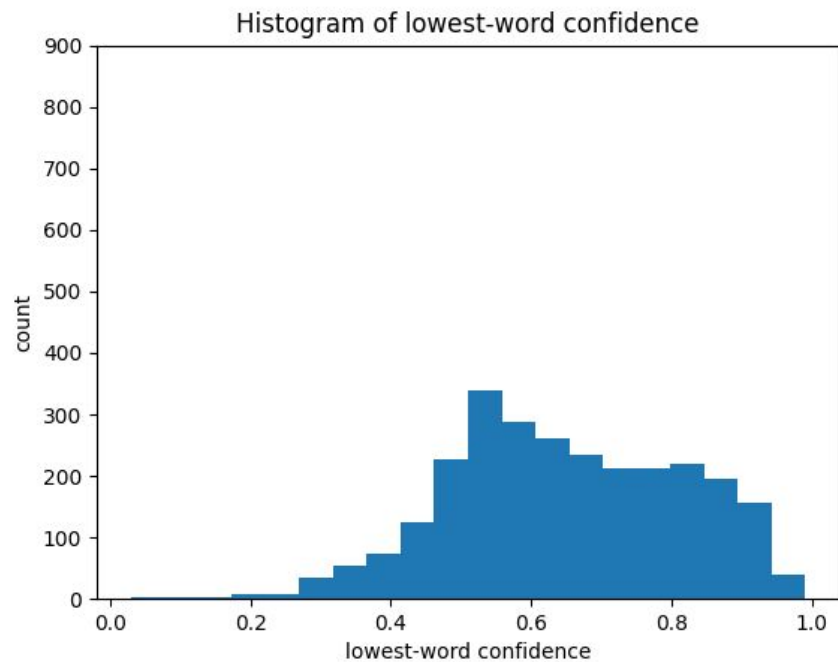
Results for gpt-4-turbo



Lowest word confidence

Second we continue with lowest word confidence level in medium whisper model and also clean Librispeech.

Histogram



Let's look at the results for GPT-3.5-turbo first:

We have two models here: gpt-3.5-turbo-1106 and gpt-3.5-turbo-0125

gpt-3.5-turbo-0125

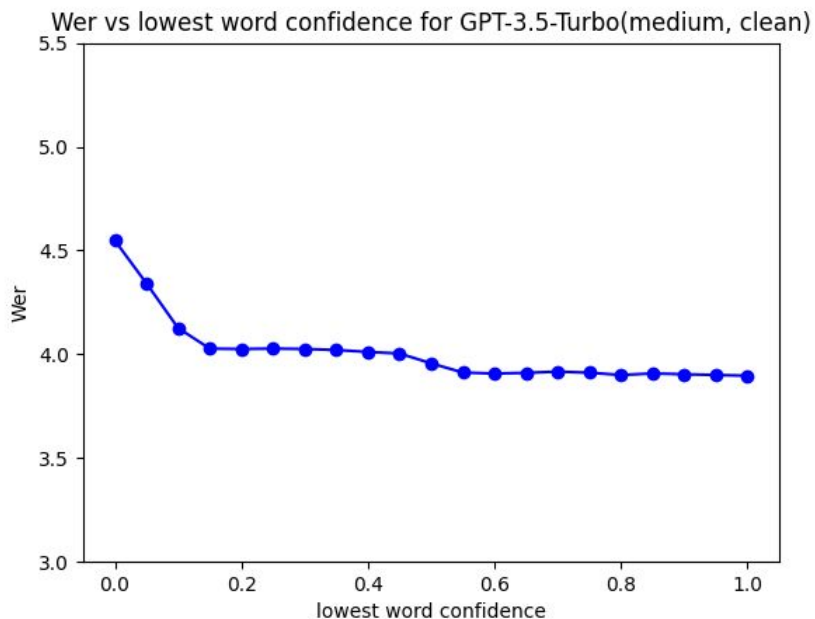
```
Number of evaluated audio files: 2702
WER_original: 4.5477%
WER_corrected_chatgpt: 3.8967%
CER_original: 1.9548%
CER_corrected_chatgpt: 1.5136%
SER_original: 39.0081%
SER_corrected_chatgpt: 38.3420%
---
```

gpt-3.5-turbo-1106

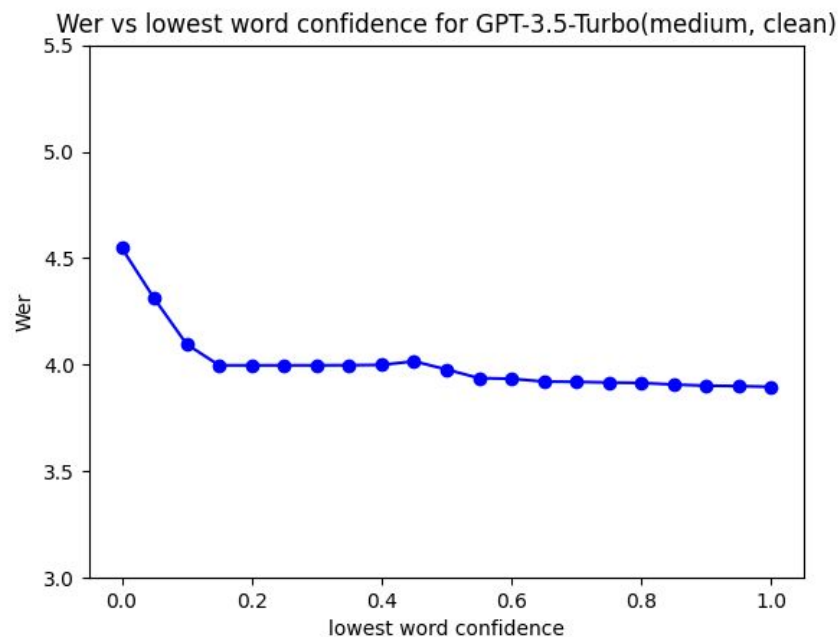
```
Number of evaluated audio files: 2702
WER_original: 4.5477%
WER_corrected_chatgpt: 3.8963%
CER_original: 1.9548%
CER_corrected_chatgpt: 1.5384%
SER_original: 39.0081%
SER_corrected_chatgpt: 38.3420%
---
```


WER vs word confidence

gpt-3.5-turbo-0125

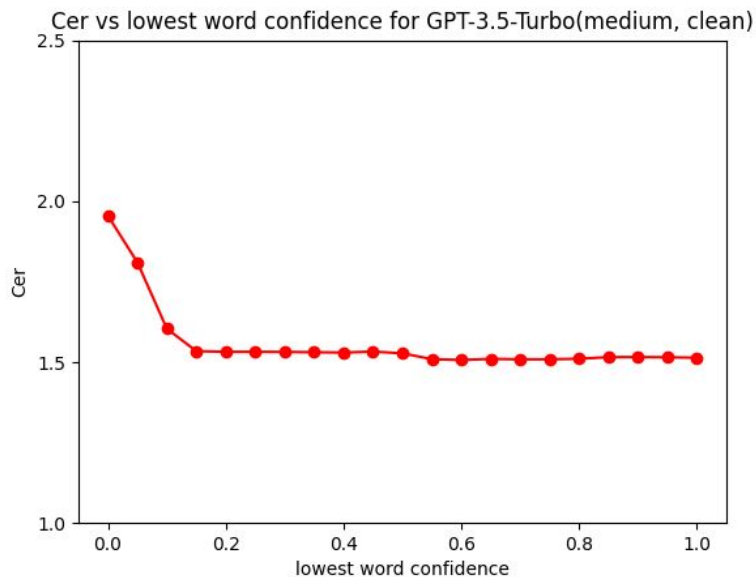


gpt-3.5-turbo-1106

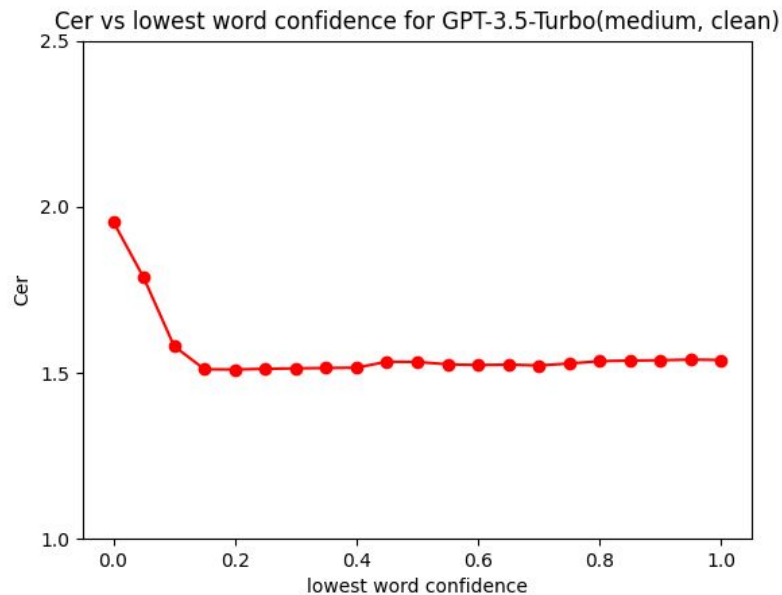


CER vs word confidence

gpt-3.5-turbo-0125



gpt-3.5-turbo-1106



Find best prompt

In this step we use gpt-3.5-turbo.

We have gpt-3.5-turbo-1106 as model.

We have two confidence level: sentence confidence level & lowest word confidence

Old prompt

You are a helpful assistant that corrects ASR errors. \

You will be presented with an ASR transcription of Librispeech data provided by the Whisper model. \

Your task is to correct any errors in the transcription. \

Provide the most probable corrected transcription in string format. \

If you come across errors in ASR transcription, make corrections that closely match the original transcription acoustically or phonetically. \

Do not change the case, for example, lower case or upper case, in the transcription. \

Do not output any additional text that is not the corrected transcription. \

Do not write any explanatory text that is not the corrected transcription.

New prompt

You are a helpful assistant that corrects ASR errors. \

You will be presented with an ASR transcription delimited by {delimiter} characters,\

and your task is to correct any errors in it. \

Provide the most probable corrected transcription in string format. \

Do not change the case, for example, lower case or upper case, in the transcription. \

Do not output any additional text that is not the corrected transcription. \

Do not write any explanatory text that is not the corrected transcription.

Finding best prompt in
sentence confidence

In sentence confidence

New prompt

```
Number of evaluated audio files: 2702
WER_original: 4.5477%
WER_corrected_chatgpt: 3.9760%
CER_original: 1.9548%
CER_corrected_chatgpt: 1.5853%
SER_original: 39.0081%
SER_corrected_chatgpt: 39.7853%
---
```

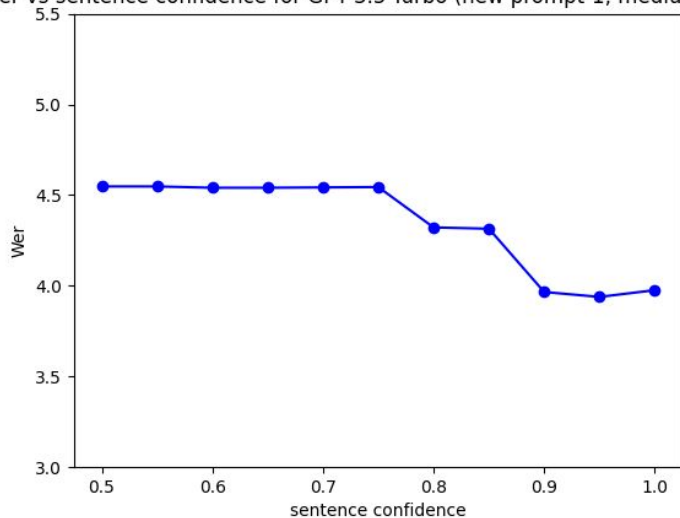
Old prompt

```
Number of evaluated audio files: 2702
WER_original: 4.5477%
WER_corrected_chatgpt: 3.9310%
CER_original: 1.9548%
CER_corrected_chatgpt: 1.5561%
SER_original: 39.0081%
SER_corrected_chatgpt: 38.3050%
---
```


WER vs sentence confidence

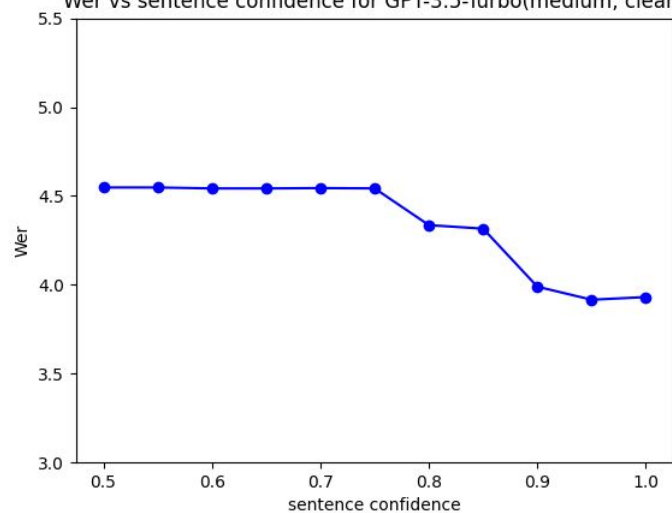
New prompt

Wer vs sentence confidence for GPT-3.5-Turbo (new prompt-1, medium, clear)



Old prompt

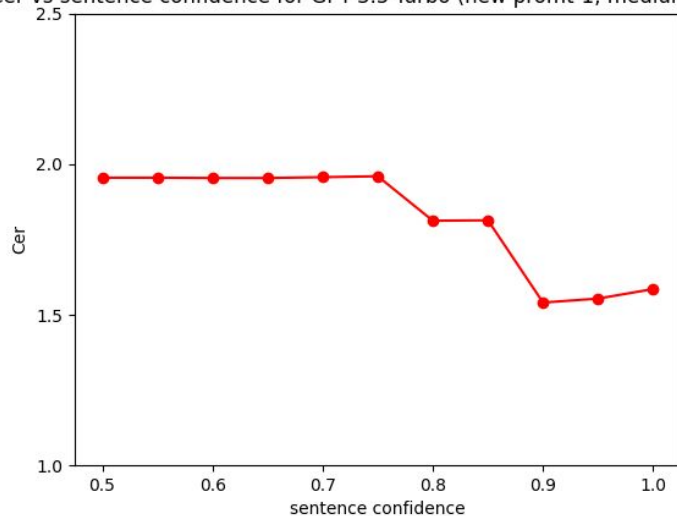
Wer vs sentence confidence for GPT-3.5-Turbo (medium, clean)



CER vs sentence confidence

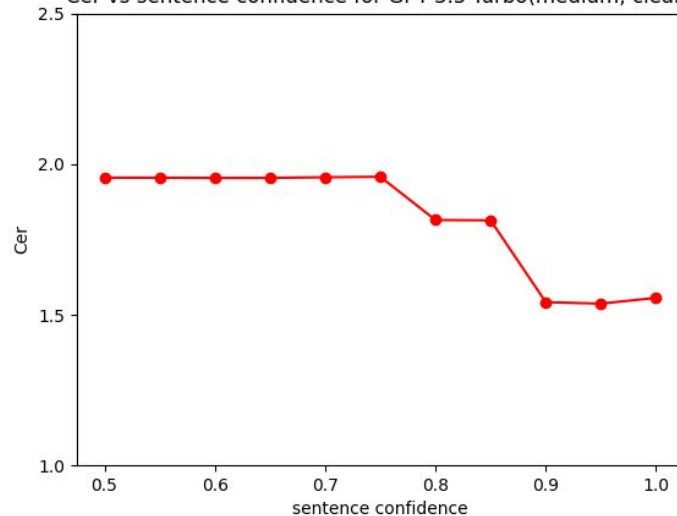
New prompt

Cer vs sentence confidence for GPT-3.5-Turbo (new prompt-1, medium, clean)



Old prompt

Cer vs sentence confidence for GPT-3.5-Turbo (medium, clean)



Finding best prompt in
lowest word confidence

New prompt

```
Number of evaluated audio files: 2702
WER_original: 4.5477%
WER_corrected_chatgpt: 4.0427%
CER_original: 1.9548%
CER_corrected_chatgpt: 1.6027%
SER_original: 39.0081%
SER_corrected_chatgpt: 40.4145%
---
```

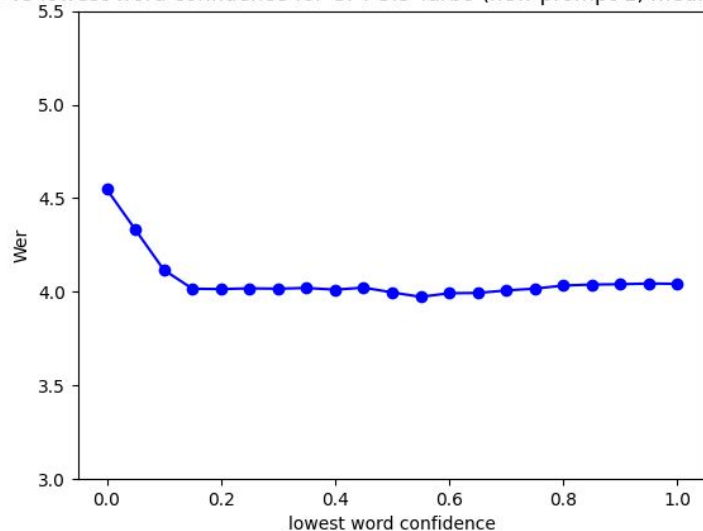
Old prompt

```
Number of evaluated audio files: 2702
WER_original: 4.5477%
WER_corrected_chatgpt: 3.8963%
CER_original: 1.9548%
CER_corrected_chatgpt: 1.5384%
SER_original: 39.0081%
SER_corrected_chatgpt: 38.3420%
---
```

WER vs lowest word confidence

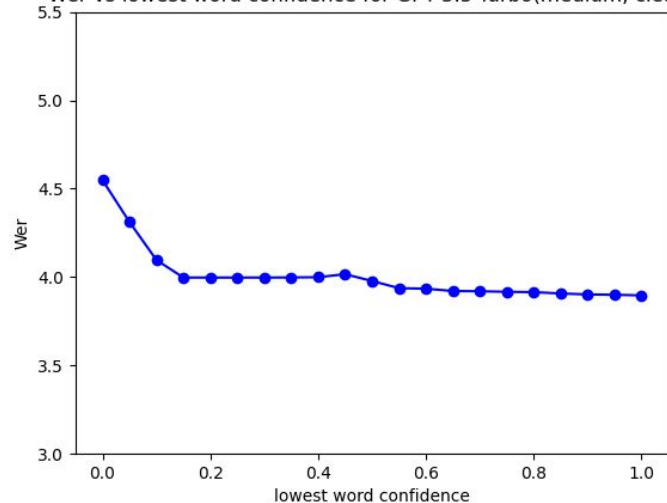
New prompt

Wer vs lowest word confidence for GPT-3.5-Turbo (new prompt-1, medium, clean)



Old prompt

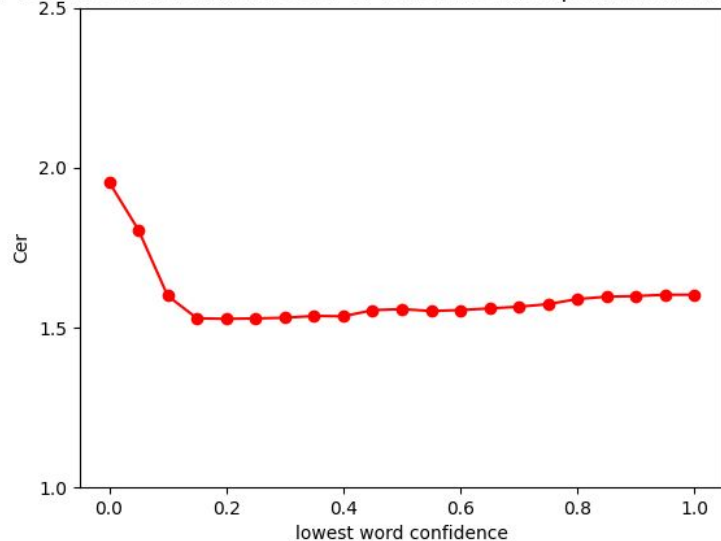
Wer vs lowest word confidence for GPT-3.5-Turbo (medium, clean)



CER vs lowest word confidence

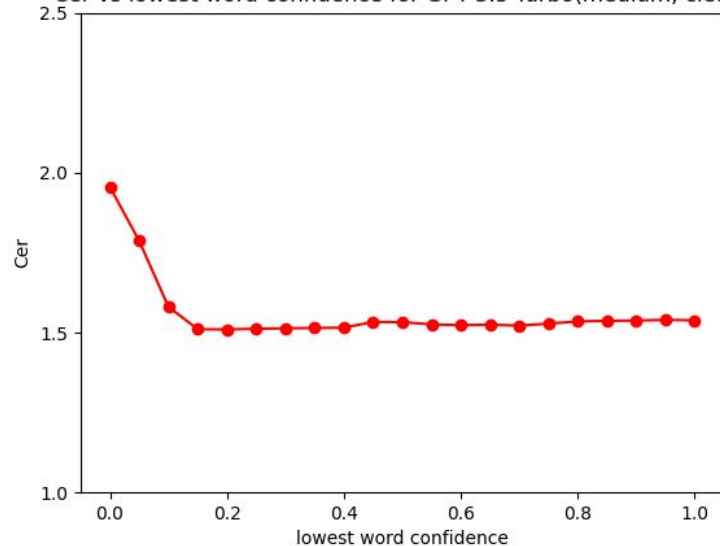
New prompt

Cer vs lowest word confidence for GPT-3.5-Turbo (new prompt-1, medium, clean)



Old prompt

Cer vs lowest word confidence for GPT-3.5-Turbo(medium, clean)



End